

	$-2.20\text{ N} - 2.2\text{ N}$
6(a)	Similarity: both quantities are defined as work done per unit charge, hence they have the same SI unit. Difference: Emf is defined as work done per unit charge converting from non-electrical to electrical energy whereas p.d. is defined as work done per unit charge converting from electrical to non-electrical energy.
6(b)(i)	Due to the internal resistance r of the cell, the voltmeter reading V will always be less than its emf when a current I flows through. $V = 1.5 - Ir$, therefore $V < 1.5$ if I and r are not zero.

Qn	Solution
6(b)(ii)	Total external resistance = $3 + \frac{(6+4) \times 10}{(6+4)+10}$ $= 3 + 5$ $= 8\Omega$ $V = IR$ $= 0.15(8)$ $= 1.2 \text{ V}$
6(c)(i)	The net external resistor R is increased, the current will decrease. ($I = E/(R+r)$) The voltmeter reading will increase as the current decreases ($V = E - Ir$)
6(c)(ii)	From b(ii) $V = E - Ir$ $1.2 = 1.5 - 0.15r$ $r = 2.0 \Omega$ From Fig 6.2 Effective resistance of parallel combination with $10\Omega = \frac{(9+6)(10)}{9+6+10}$ $= 6\Omega$ Pd across this parallel combination $= \frac{6}{6+3+2} E$ $= \frac{6(1.5)}{11}$ $= 0.818 \text{ V} = 0.82 \text{ V}$
7(a)	The magnitude of the induced emf in a conductor is proportional to the rate of